

20250616EB_MN (NCT, ATP, TDP43)

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2025年6月16日 星期一

Differentiation: BP w/ 1.5mM pyruvate.

Experiment: NB-A with 1mM pyruvate, N2, B27, Glutamax, T8 supps

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2025年6月18日 星期三

T2 -> T3

2025年6月20日 星期五

T3 -> T4 (~14 mL for over-weekend)

2025年6月23日 星期一

T4 -> T5

2025年6月25日 星期三

T5 -> T6

2025年6月27日 星期五

T6 -> T6 (~14mL for over-weekend)

2025年6月30日 星期一

T6 -> T7

Dissociation (modify the protocol a bit and try trypsin and accutase method (advice from Dr. Ya-Ping) in parallel)

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) o/n
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine o/n
4. Collect EBs in 50mL falcon and spin them down with 100rcf-200ref, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs). (We can still see a bit EBs are floating, but most of them are spun down)
5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)

- a. We do 12 min in total for trypsin this time.

b. For accutase: we incubate the cells in 37 and flick the tube around every 5 min until it gets a bit cloudy. I did 40 min in total this time.
7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 100rcf, 3.5min ~~300rcf, 7min~~. Remove the supernatent. (We can still see some clumps in the supernatent. When we span down the supernatent at 200rcf, 2min and checked the supernatent directly, we see lots of big chunk in trypsin group along with many single cells. For accutase group, we see many cells in single cells or clusters, but not many huge chunk)
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

coating date					
	A	B	C	D	E
1		A	B	C	D
2	0.01% PLO with Na borate	2025/6/27	2025/6/13	2025/6/6	2025/6/6
3	laminin	2025/6/28	2025/6/14	2025/6/7	2025/6/7

cell info			
	A	B	C
1	cell	passage	well#
2	WT11	16+19	6 *2

Table1		
	A	B
1	plate1	volume (mL)
2	trypsin	3
3	HI-FBS/CTWM	6
4	plate2	
5	accutase	4
6	HI-FBS/CTWM	8

Table3									
	A	B	C	D	E	F	G	H	I
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	1.25	96	5	6.25	A	24	250	12	2400
3					C	96	60	24	1152
4	4.78	88	6	28.68	A	24	250	12	667
5	4.5	<- use this instead since there are many non-single cell clumps			B	24	250	14	778
6					D	24	250	4	222

Plate A						
	1	2	3	4	5	6
A				92 5uL/4 well	110 12uL/ 4well	111 ~8- 10uL/ 4well
B						
C						
D						

Plate B						
	1	2	3	4	5	6
A	92 5uL/4 well			no virus		no virus
B						
C						
D						

Plate C													^
	1	2	3	4	5	6	7	8	9	10	11	12	
A													
B													
C													
D													
E													
F													
G													
H													

Plate D							^
	1	2	3	4	5	6	
A						new 110 1uL	
B						new 110 3uL	
C						new 111 1uL	
D						new 111 3uL	

2025年7月1日 星期二

Change T7 -> T7 if added virus

2025年7月3日 星期四

T7 -> BP T8

almost all cells on plate C died -> only around 8 wells survived. It might because of the coating or the luminascence reagent.
Since almost wells far from the used one survived.

Plate C1												^
	1	2	3	4	5	6	7	8	9	10	11	12
A												
B										some survived		
C												
D												
E										died		
F												
G												
H												

2025年7月4日 星期五

add extra 500uL BP T8 (Help from Stephanie)
(500uL*38 = 16mL, 100uL*24 = 2.4mL. In total: ~19mL)

2025年7月7日 星期一

-700 BP T8 -> +300 BP T8 (Help from Stephanie)

2025年7月8日 星期二

Check Plate D (check which virus volume is better)

1ul has expression, but the intensity is in general lower
3ul has more cells expressing mCherry and has more higher intensity cells. (use 3ul/well for both virus in the future)

2025年7月9日 星期三

1/2 BP T8 -> 1/2 BP T8 (those are not imaged)

Image plate A (NCT, ATP) & B (ATP)

Imaging (NCT, column5&6)

Plate A conditions							^
	1	2	3	4	5	6	
A				1	1	1	
B				2	2	2	
C				3	3	3	
D				4	4	4	

Try the condition 7 vs. 8 from 20250527MN

condition from 20250527MN													
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	condition	media	pyruvate	N2(100x), B27 (50x)	N2	B27	GlutaMax (100x)	Glutamax	T8 supp	glucose	1.1M glucose	drug	volume
2		7	NB-A	1mM	v	20	40	v	20	v	2.5	4.55	2000
3		8	NB-A	1mM	v	20	40	v	20	v	0		2000

Prepare more media (30mL)

condition from 20250527MN1													
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	media	pyruvate (100x)	100mM pyruvate	N2(100x), B27 (50x)	N2	B27	GlutaMax (100x)	Glutamax	T8 supp	glucose	1.1M glucose	drug	volume (uL)
2	NB-A	1mM	300	v	300	600	v	300	v	0			30000

Prepare media as below

media prep											
	A	B	C	D	E	F	G	H	I	J	
1											
2	condition #	glucose conc	diluted from (mM)	glucose volume	total volume (uL)	drug	volume w/ SirDNA	SirDNA (1:5000)	volume w/ JFX650 & SPY595	JFX650 & SPY595 (1:2500)	
3	1	2.5	1100	20.45	9000		2000	0.4	800	0.32	
4	2	0.15	2.5	300.00	5000		2000	0.4	800	0.32	
5	3	0		0	5000		2000	0.4	800	0.32	
6	4	0		0	3000	FCCP (3uL), 2DG (9uL)	2000	0.4			

prepare media with different conc. of 2DG in 2.5mM glu NB-A media							
	A	B	C	D	E	F	G
1	condition #	diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)	media volume
2	5	2000	6	333	800	2.4	797.6
3	6	2000	2	1000	800	0.8	799.2
4	7	6	0.6	10	800	80.0	720.0
5	8	6	0.2	30	800	26.7	773.3

Treatment time (all)															
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1				110 (plate A)		111 (plate A)		92 (plate A)		92 (plate B)		TDP43 (plate B)			
2	plate	#		treat time	image time	treat time	image time	treat time	image time	treat time	image time	treat time	treat JFX, SPY	image time	
3	A	1	2.5mM glu	13:00	14:02	10:17	11:25								
4		2	0.15mM glu	13:40	15:02	10:45	11:52								
5		3	0mM glu	14:20	15:30	11:23	12:25								
6		4	0mM glu+FCCP+ 2DG	15:20	16:20	12:08	13:08								
7	B	5	6mM 2DG							15:50	17:00				
8		6	2mM 2DG							16:05	17:07				
9		7	0.6mM 2DG							16:20	17:20				
10		8	0.2mM 2DG							16:35	17:42				
11	A/B	1	2.5mM glu					17:00	18:00			17:25	16:40	17:50	
12		2	0.15mM glu					17:15	18:23			17:25	16:40	18:30	* laser didn't work at the beginning...
13		3	0mM glu					17:30	18:33			17:25	16:40	18:40	
14		4	0mM glu+FCCP+ 2DG					17:45	18:43						
15		7/10													
16	B	1	2.5mM glu										16:10	17:12	
17		2	0.15mM glu										16:20	17:27	
18		3	0mM glu										16:30	17:40	

NCT_O.C. , 60X, 600x900, 16-bit					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	16	30	300 ms	
3	640	16	10	100 ms	
4	488	16	50	1s	

10 positions/well (10 positions would be the max. there is only 2 sec between loops)

Used 16 bit instead of 12 bit (the S/N is much better!!!!)

110: pre-activation: 3 loops, 30s interval; activation: 30 loops, 30s interval

111: pre-acitvation: 3 loops, 30s interval; activation: 25 loops, 30s interval

*note: I didn't set correct O.C. for 488 for 111 (conditin 1). I stop the acquisition at the first round of activation, so some of the cells had exposed to 488 once. I waited a couple of minute before starting a new round.

Image (ATP/ADP ratio with vLH92, column4)

O.C. , 60X, 600x900, 16-bit					^
	A	B	C	D	
1	channel	bit	power (%)	exposure time	
2	488	16	20	200	
3	Dapi-ex-Green-em	16	20	300	

10positions/well
Treat cells for an hour and image the end point ATP/ADP ratio

2025年7月10日 星期四

Image Plate B (TDP43)

Plate B1							^
	1	2	3	4	5	6	
A	5			1 24h	2 24h	3 24h	
B	6					1	
C	7					2	
D	8					3	

Treat cells with 2.5uM JFX650 and SPY595 (1:2500) 1h before the imaging.

O.C. , 60X, 600x900, 16-bit1						^
	A	B	C	D	E	
1	channel	bit	power (%)	exposure time	note	
2	561	16	20	500ms		
3	640	16	10	300ms		
4	640 movie	16	40	500ms	40 loops	

Imaging:

~10 positions for snapshot (odd fileids)
6 positions for neurites (even fileids)